# Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller

Software Classification

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|---|---|
| Reference | TBD **[TBD-1]** |
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| Date | 11 October 2026 |
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| Open items | 1 TBD (App. A) |

**Document change record**

| Issue | Rev. | Date | Change |
|---|---|---|---|
| 1 | 0 | 2026-10-10 | First draft |
| 1 | 1 | 2026-10-11 | Safety-critical designation from the hazard analysis |

## 1.0 Purpose and Scope

This document establishes the software classification of each software item, and so the rigour each item requires. It is the formal record of the software classification determinations [1, Sec. 3.5.2, p. 25].

The evaluated software items encompass the System software and its enabling systems, as defined in the Concept of Operations [4, Sec. 1.2.2]. The System models an uncrewed spacecraft whose primary mission objectives directly depend on this software; a failure would cost those objectives.

## 2.0 Documents

[1] NASA, *NASA Software Engineering Requirements*, NPR 7150.2D.
[2] NASA, *Software Assurance and Software Safety Standard*, NASA-STD-8739.8B, 2022.
[3] NASA, *NASA Software Engineering Handbook*, NASA-HDBK-2203, Version D, page "SWE-020 – Software Classification".
[4] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Concept of Operations*, Issue 1, Revision 1, Draft.
[5] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Hazard Analysis*, Issue 1, Revision 0, Draft.

## 3.0 Classification

Each item takes the highest class whose definition applies [1, Sec. 3.5.1, p. 24].

| Software item | Role | Class |
|---|---|---|
| Controller firmware | FLIGHT | B |
| Onboard computer software | FLIGHT | B |
| Ground software in the mission operations center | GROUND | B |
| Bench test tools | TEST | C |
| Firmware build tools | BUILD | D |

**Legend: role**

| | |
|---|---|
| **FLIGHT** | Operates the spacecraft from orbit. |
| **GROUND** | Operates the spacecraft from the ground. |
| **TEST** | Tests the System; never operates the spacecraft. |
| **BUILD** | Builds the System; never operates the spacecraft. |
| **Class** | As defined in [1, App. D, pp. 79–86]. |

***Note.*** *The class decides which rules of NPR 7150.2D apply to each item [1, Sec. 3.1.11, p. 22]: all 100 rules apply to Class B, 92 to Class C and 64 to Class D [1, App. C, pp. 56–78]. Appendix B records which of them the project follows.*

### 3.1 Rationale

**Controller firmware: Class B.** Class B covers flight software "required to operate the vehicle or space asset ... such as commanding of the vehicle or asset" [1, App. D, p. 81]; its examples include "fault protection" [1, App. D, p. 81]. The controller runs commands and returns to working firmware on its own when new firmware fails.

**Onboard computer software: Class B.** The same definition applies. It passes every command and firmware image to the controller, watches it, and restores it when its firmware cannot run.

**Ground software: Class B.** Class B examples include "command and control ground systems" [1, App. D, p. 81]. The ground software prepares and checks every command and firmware image sent to the spacecraft.

**Bench test tools: Class C.** Class C covers "software used for the testing of space assets" [1, App. D, p. 82], and Class B excludes systems "used in testing Class B systems" in a development environment [1, App. D, p. 81].

**Firmware build tools: Class D.** Class D covers "ground software tools that support engineering development" [1, App. D, p. 83].

## 4.0 Safety-Critical Designation

Some software is safety-critical: the parts that can cause a hazard, or that exist to prevent, detect or handle one [2, Sec. 4.2, p. 20]. Safety-critical software must meet extra requirements on top of its class, for example test coverage of every decision (MC/DC) [1, Sec. 3.7.4, pp. 26–27].

The hazard analysis designates the items below from its hazards [5, Sec. 4.0]. Until the architecture divides each item into components, the whole item is treated as safety-critical.

| Software item | Safety-critical | Criteria | Hazards |
|---|---|---|---|
| Controller firmware | YES | a, b, c, e | H-1 to H-4, H-6 to H-13 |
| Onboard computer software | YES | a, b, c, e | H-1 to H-17 |
| Ground software | YES | a, b | H-2, H-4 to H-7, H-9, H-10, H-15, H-17 |
| Bench test tools | NO | NONE | NONE |
| Firmware build tools | NO | NONE | NONE |

**Legend: criteria** [2, Sec. 4.2, p. 20]. a: causes or contributes to a system hazardous condition or event. b: controls functions identified in a system hazard. c: provides mitigation for a system hazardous condition or event. d: mitigates damage if a hazardous condition or event occurs. e: detects, reports and takes corrective action if the system reaches a potentially hazardous state. Hazards: the hazard IDs of [5, Sec. 4.0].

***Note.*** *Bench test tools stay out of operation (H-12). Firmware build tools never run in operation; a defect they put into firmware is a shared cause [5, Sec. 5.2].*

## 5.0 Tailoring and Review

- **Joint agreement and single-developer tailoring:** Software engineering and software assurance are expected to agree on each classification [1, Sec. 3.5.1 Note, p. 25]. One engineer develops the System, with no independent software assurance team (ConOps constraint C-3); the classification therefore rests with the author alone.
- **Life-cycle re-evaluation:** The classification is revisited at each later phase [3, Sec. 3.5.1]. If an item's class changes, the plans are updated to match [1, Sec. 2.2.8, p. 19].

## A Open Items

The values below are not yet known.
**TBD-1** Project name and document reference.
The author accepted the safety-critical designation (Section 4.0) on 11 October 2026.

# B Requirements Mapping Matrix

NPR 7150.2D sets 100 rules for how software is built, each marked for the classes it applies to [1, App. C, pp. 56–78]. The project intends to follow every rule that applies to an item's class, except the 20 tailored out in the second table below [1, Sec. 3.1.12, p. 22]. The author decided this tailoring on 10 October 2026; it is revisited at each later phase.

| Class | Rules that apply | Tailored out | Followed |
|---|---|---|---|
| B | 100 | 20 | 80 |
| C | 92 | 15 | 77 |
| D | 64 | 11 | 53 |

**Followed by the Class B items** (by NASA's rule number, SWE): 013, 016, 017, 018, 020, 023, 024, 027, 033, 034, 036, 037, 050, 051, 052, 053, 054, 055, 057, 058, 060, 061, 062, 063, 065, 066, 068, 071, 073, 075, 077, 079, 080, 081, 082, 083, 084, 085, 086, 087, 088, 089, 090, 093, 121, 125, 134, 135, 136, 139, 146, 147, 154, 156, 157, 159, 176, 184, 185, 186, 187, 189, 190, 191, 192, 193, 194, 195, 196, 199, 200, 201, 202, 203, 204, 207, 210, 211, 219, 220. The Class C and D items follow those of these rules marked for their class.

| SWE | Requirement | Reason |
|---|---|---|
| 141 | Software IV&V for NASA project categories | NOT NASA |
| 143 | Architecture review for NASA project categories | NOT NASA |
| 174 | Planning data to the Center measurement repository | NOT NASA |
| 148 | Reuse candidates to NASA's catalogue | NOT NASA |
| 094 | Measurement data to the sponsoring organisation | NOT NASA |
| 039 | Status reports from the software developer | NO SUPPLIER |
| 040 | Products and records from the developer to NASA | NO SUPPLIER |
| 042 | NASA access to the developer's source code | NO SUPPLIER |
| 046 | Software schedule from the developer | NO SUPPLIER |
| 206 | NASA access to the developer's models and simulations | NO SUPPLIER |
| 045 | Joint NASA/developer audits | NO SUPPLIER |
| 131 | IV&V plan | NO IV&V |
| 178 | IV&V access to development artifacts | NO IV&V |
| 179 | Responses to IV&V issues and risks | NO IV&V |
| 032 | CMMI-DEV rated organisation | NO RATING |
| 022 | Software assurance and safety plan per NASA-STD-8739.8 | NO ASSURANCE |
| 205 | Safety-critical determination with the SMA organisation | NO ASSURANCE |
| 015 | Two cost estimate models | NO BUDGET |
| 151 | Conditions on the cost estimate | NO BUDGET |
| 070 | Accredited models and simulations for flight qualification | BENCH ONLY |

**Legend: reason**

**NOT NASA** The System is not a NASA program or project; there is no NASA Center, catalogue or sponsoring organisation.

**NO SUPPLIER** The engineer develops all software; there is no separate developer or supplier.

**NO IV&V** Independent verification and validation is not performed.

**NO RATING** There is no organisation to hold a CMMI rating.

**NO ASSURANCE** There is no independent software assurance or SMA organisation (ConOps constraint C-3); the author decides alone.

**NO BUDGET** The project has no development budget to estimate.

**BENCH ONLY** The System is demonstrated on a bench, not qualified for flight (ConOps constraint C-4).